

Diameter Server — application.yaml

Environment Variable Reference & Configuration Guide

This document describes every externalized environment variable defined in the Diameter Server's application.yaml, along with its default value and purpose. Variables are grouped by functional area: Server, Database, Logging, Service Discovery, Application Info (Diameter / stack / audit), Audit CSV Writer, Audit Routing, and Messaging. Fixed (non-externalized) properties are listed separately for completeness.

1. Server Configuration

Controls the embedded web server port used for Actuator and REST endpoints.

Environment Variable	Default Value	Description
ENV_SERVER_PORT	8080	TCP port on which the embedded web/application server listens for HTTP requests (Actuator, REST APIs).

2. Application Identity

Defines how the service registers itself and appears in logs and metrics.

Environment Variable	Default Value	Description
ENV_SERVER_NAME	diameter-server	Logical application name used for service identification, Eureka registration, and log/metric labelling.

3. Database Configuration (MySQL / HikariCP)

JDBC connection details and HikariCP connection-pool tuning parameters.

Environment Variable	Default Value	Description
ENV_DB_URL	<code>jdbc:mysql://216.48.180.92:4406/diameter-sit-db</code>	JDBC connection URL for the MySQL database — includes host, port, and schema name.
ENV_DB_USERNAME	<code>root</code>	Username for MySQL authentication.
ENV_DB_PASSWORD	<code>diameterserver@2024@db</code>	Password for MySQL authentication. Must always be injected via a secrets manager in production; never hard-coded.
ENV_MAX_DB_POOL_SIZE	<code>100</code>	Maximum connections HikariCP may hold (idle + in-use). Tune against MySQL <code>max_connections</code> and peak session load.
ENV_MIN_DB_IDLE_POOL_SIZE	<code>20</code>	Minimum idle connections HikariCP keeps warm so they are immediately available under load.
ENV_DB_CONNECTION_TIMEOUT	<code>3000 ms</code>	Maximum time a caller waits for a pool connection before an exception is thrown.
ENV_DB_IDLE_TIMEOUT	<code>300000 ms</code>	Maximum time a connection may sit idle in the pool (above minimum-

Environment Variable	Default Value	Description
		idle) before being retired.
ENV_DB_MAX_LIFETIME	1800000 ms	Maximum lifetime of any connection in the pool before it is retired and replaced, avoiding stale connections.
ENV_DB_LEAK_DETECTION_THRESHOLD	60000 ms	If a connection is borrowed from the pool for longer than this duration it is logged as a possible leak.

Note: data-source-properties.rewriteBatchedStatements is fixed to true to improve batch INSERT/UPDATE performance with MySQL and is not externalized.

4. Logging Configuration

Log-level controls per category. Log configuration file is fixed: classpath:log4j2-spring.xml.

Environment Variable	Default Value	Description
ENV_STACK_LOG_LEVEL	INFO	Log verbosity for the STACK logger category (framework/stack layer). Raise to DEBUG/TRACE for troubleshooting.
ENV_DIAMETER_LOG_LEVEL	INFO	Log verbosity for the com.diameter package (application Diameter logic).
ENV_ROOT_LOG_LEVEL	INFO	Fallback log level applied to all loggers that do not have a more specific level configured.

5. Service Discovery (Eureka)

Registration and discovery settings for the Eureka service registry. *instance-id* is derived from application name + port; *prefer-ip-address* is fixed to *true*.

Environment Variable	Default Value	Description
ENV_EUREKA_URL	http://216.48.180.92:8761/eureka/	Default-zone URL of the Eureka service registry. Must point to the correct environment-specific registry when promoting builds.

6. Application Info & Diameter Stack Configuration

Custom metadata exposed via */actuator/info* and used to configure Diameter protocol behavior, thread pool sizing, cache management, chunking, audit enablement, and Rx interface identity.

Environment Variable	Default Value	Description
ENV_IDENTITY	diameter.com	Diameter Origin-Host / stack identity used by this server to identify itself in Diameter protocol AVPs.
ENV_REALM	local	Diameter realm under which this server operates, used in Origin-Realm AVPs.
ENV_DIAMETER_SERVER_CONFIGURATION	TCP	Transport protocol for Diameter peer connections. Supported values: TCP, SCTP.
ENV_QUOTA_UNIT	BYTES	Unit of measurement for data quota tracking and reporting (e.g. BYTES, KILOBYTES).
ENV_TIME_UNIT	SECOND	Unit of measurement for time-based quota tracking and session durations (e.g. SECOND, MINUTE).
ENV_GRACE_PERIOD	30	Grace period (in the configured time unit) allowed after quota exhaustion or plan expiry before enforcement action.
ENV_PLAN_EXPIRY_CHECK_IN_RX	false	Boolean flag — when true, plan expiry is evaluated

Environment Variable	Default Value	Description
		during Rx interface interactions (AF/PCRF).
ENV_POD_NAME	diameter_server	Name of the running pod/instance, exposed via the Actuator /info endpoint for cluster identification.
ENV_STACK_REQUEST_TIMEOUT	5000 ms	Maximum time the Diameter stack waits for a response to an outbound request before timing out.
ENV_STACK_MIN_THREAD_POOL_SIZE	50	Minimum number of threads kept ready in the Diameter stack worker thread pool.
ENV_STACK_MAX_THREAD_POOL_SIZE	200	Maximum threads the Diameter stack worker pool can grow to under sustained load.
ENV_THREAD_KEEP_ALIVE_TIME	300000 ms	Time an idle thread above the minimum pool size is kept alive before it is terminated.
ENV_MAX_REQUEST_QUEUE_SIZE	50000	Maximum pending requests that can be queued before the stack starts backpressuring new requests.
ENV_CACHE_CLEANUP_INTERVAL	300000 ms	Interval at which in-memory caches (session, quota) are swept and expired entries removed.
ENV_LIVE_SESSION_EXPIRY	7200 sec	Time after which an inactive live Diameter session is considered expired and eligible for cleanup.
ENV_MAX_CHUNK_SIZE_MB	100 MB	Maximum data volume of a single processing chunk before it is split or flushed.
ENV_MAX_CHUNK_SIZE_SECOND	120 sec	Maximum time window a processing chunk may span before it is closed/flushed, working alongside the MB limit.
ENV_AUDIT_ENABLE	true	Master switch for all audit logging. When false, no

Environment Variable	Default Value	Description
		audit records are written regardless of routing configuration.
ENV_RX_ORIGIN_HOST	rxasbc83.asbc.system.com	Diameter Origin-Host identity used specifically on the Rx interface (AF ↔ PCRF). Allows the Rx peer to identify itself with a distinct host identity separate from the main stack identity (ENV_IDENTITY).

7. Audit Configuration

Controls how Diameter request/response audit records are persisted — as database rows and/or rotating CSV files — with independent per-request-type routing. All routing is skipped when ENV_AUDIT_ENABLE is false.

7.1 Audit CSV Writer Settings

Environment Variable	Default Value	Description
ENV_AUDIT_CSV_DIRECTORY	D:/diameter-audit	Filesystem path where audit CSV files are written. Override with a Linux path (or mounted volume path) in container/Kubernetes deployments.
ENV_AUDIT_CSV_QUEUE_SIZE	100000	Capacity of the in-memory audit record queue. When the queue is full, new records may be dropped or backpressured.
ENV_AUDIT_CSV_BATCH_SIZE	500	Number of audit records grouped into a single batch write to the CSV file, improving write efficiency.
ENV_AUDIT_CSV_FLUSH_INTERVAL_MS	5000 ms	Maximum interval between flushes of buffered audit records to disk even if a full batch has not yet accumulated.
ENV_AUDIT_CSV_ROTATE_SIZE_MB	500 MB	File size at which the current audit CSV file is rotated and a new file is started.
ENV_AUDIT_CSV_BUFFER_SIZE	1048576 bytes	Size of the internal I/O write buffer used when appending audit records to CSV, affecting write throughput.

7.2 Audit Routing by Request Type

Each request type independently routes audit records to the database (db) and/or CSV file (csv). Both flags can be true simultaneously.

Environment Variable	Default Value	Description
ENV_INITIAL_REQUEST_AUDIT_ROUTING_DB	true	Route INITIAL_REQUEST audit records to the database.
ENV_INITIAL_REQUEST_AUDIT_ROUTING_CSV	false	Route INITIAL_REQUEST audit records to the CSV audit file.
ENV_UPDATE_REQUEST_AUDIT_ROUTING_DB	false	Route UPDATE_REQUEST audit records to the database.
ENV_UPDATE_REQUEST_AUDIT_ROUTING_CSV	true	Route UPDATE_REQUEST audit records to the CSV audit file.
ENV_TERMINATION_REQUEST_AUDIT_ROUTING_DB	true	Route TERMINATION_REQUEST audit records to the database.
ENV_TERMINATION_REQUEST_AUDIT_ROUTING_CSV	false	Route TERMINATION_REQUEST audit records to the CSV audit file.
ENV_EVENT_REQUEST_AUDIT_ROUTING_DB	false	Route EVENT_REQUEST audit records to the database.
ENV_EVENT_REQUEST_AUDIT_ROUTING_CSV	true	Route EVENT_REQUEST audit records to the CSV audit file.
ENV_UNKNOWN_REQUEST_AUDIT_ROUTING_DB	false	Route UNKNOWN_REQUEST (unrecognised command) audit records to the database.
ENV_UNKNOWN_REQUEST_AUDIT_ROUTING_CSV	true	Route UNKNOWN_REQUEST audit records to the CSV audit file.

8. Kafka Configuration

Kafka broker connectivity and consumer/producer tuning. Only the broker address is externalized; the remaining settings are fixed and injected via @Value.

8.1 Externalized via Environment Variable

Environment Variable	Default Value	Description
<code>KAFKA_URL</code>	<code>216.48.180.92:30084</code>	Bootstrap server address (host:port) of the Kafka broker for producer/consumer connections.

8.2 Fixed Kafka / Consumer / Producer Properties

Hard-coded in application.yaml. Listed here for operational visibility.

Environment Variable	Default Value	Description
<code>spring.kafka.admin.auto-create</code>	<code>false</code>	Disables automatic topic creation; topics must be provisioned explicitly. Fixed, not externalized.
<code>kafka.max.request.size</code>	<code>2000000000</code>	Maximum size (bytes) of a single Kafka producer request, accommodating very large payloads. Fixed, not externalized.
<code>max.poll.records</code>	<code>15000</code>	Maximum records returned in one consumer poll() call. Fixed, not externalized.
<code>max.poll.interval.ms</code>	<code>300000 ms</code>	Maximum delay between consecutive poll() calls before the consumer is considered failed and triggers a rebalance. Fixed, not externalized.
<code>fetch.max.bytes</code>	<code>524288000</code>	Maximum data (bytes) the consumer fetches per request across all partitions. Fixed, not externalized.
<code>fetch.min.bytes</code>	<code>1</code>	Minimum data (bytes) the broker returns per fetch request; value 1 means respond as soon as any data is available. Fixed, not externalized.

9. Fixed JPA & Liquibase Properties

Not externalized. These settings govern schema-management behavior at startup.

Environment Variable	Default Value	Description
<code>spring.jpa.hibernate.ddl-auto</code>	none	Disables Hibernate DDL generation; schema changes are managed exclusively by Liquibase. Fixed, not externalized.
<code>hibernate.dialect</code>	<code>org.hibernate.dialect.MySQL8Dialect</code>	Hibernate SQL dialect for MySQL 8 optimized SQL generation. Fixed, not externalized.
<code>spring.jpa.show-sql</code>	false	Disables SQL statement logging to the console/log. Fixed, not externalized.
<code>spring.liquibase.change-log</code>	<code>classpath:db/changelog/changelog-master.xml</code>	Path to the Liquibase master changelog that drives schema migrations on startup. Fixed, not externalized.
<code>spring.liquibase.enabled</code>	true	Enables Liquibase to run schema migrations automatically at startup. Fixed, not externalized.

10. Fixed Actuator / Management Properties

Not externalized. Controls endpoint exposure and metrics scraping.

Environment Variable	Default Value	Description
<code>management.endpoints.web.exposure.include</code>	*	Exposes all Actuator endpoints over HTTP. Fixed, not externalized.
<code>management.endpoint.health.show-details</code>	always	Always returns full component detail in <code>/actuator/health</code> responses. Fixed, not externalized.

Environment Variable	Default Value	Description
<code>management.metrics.export.prometheus.enabled</code>	<code>true</code>	Enables Prometheus metric scraping via <code>/actuator/prometheus</code> . Fixed, not externalized.

11. Usage Notes & Recommendations

- Every `${ENV_VAR:default}` placeholder falls back to its default when the variable is not set, keeping the service runnable in local/dev without additional configuration.
- `ENV_DB_PASSWORD` and all credentials should be injected via a secrets manager, Kubernetes Secret, or CI/CD secret variable in staging and production — never committed in plain text.
- Pool sizing (`ENV_MAX_DB_POOL_SIZE`, `ENV_MIN_DB_IDLE_POOL_SIZE`) should be tuned against expected concurrent Diameter session load and the MySQL server's `max_connections` limit.
- `ENV_EUREKA_URL` and `KAFKA_URL` must be overridden per environment (dev / SIT / UAT / prod) when promoting builds; they default to SIT infrastructure addresses.
- Log level variables (`ENV_*_LOG_LEVEL`) can be raised to `DEBUG` or `TRACE` temporarily during troubleshooting and should be reverted to `INFO` in production to control log volume.
- `ENV_STACK_MIN_THREAD_POOL_SIZE`, `ENV_STACK_MAX_THREAD_POOL_SIZE`, and `ENV_MAX_REQUEST_QUEUE_SIZE` should be tuned together to handle peak Diameter request concurrency without thread starvation or queue overflow.
- `ENV_LIVE_SESSION_EXPIRY` and `ENV_CACHE_CLEANUP_INTERVAL` directly affect memory usage. Setting expiry too high without a sufficiently frequent cleanup interval can allow stale sessions to accumulate.
- `ENV_AUDIT_CSV_DIRECTORY` defaults to a Windows-style path. This must be overridden with a Linux path and a writable, persistent volume mount in container or Kubernetes deployments.
- `ENV_AUDIT_CSV_QUEUE_SIZE`, `ENV_AUDIT_CSV_BATCH_SIZE`, and `ENV_AUDIT_CSV_FLUSH_INTERVAL_MS` govern audit write throughput and crash-durability. A larger queue and longer flush interval increase throughput but risk losing more unflushed records on an unclean shutdown.
- The per-request-type audit routing flags (`ENV_*_AUDIT_ROUTING_DB / _CSV`) take effect only when `ENV_AUDIT_ENABLE` is true. Review and align routing defaults per environment, since the defaults intentionally split traffic between DB and CSV differently for each request type.
- `ENV_RX_ORIGIN_HOST` configures a distinct Diameter identity for the Rx interface. Ensure this matches the Origin-Host value expected by the PCRF or AF peer on the Rx side; a mismatch will cause Diameter peer connection rejection.